

LFM2-8B-A1B: A Mixture of Expert for Edge Intelligence: Run Locally



Fahd Mirza
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This video locally installs LFM2-8B-A1B is the on-device MoE.

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https://huggingface.co/LiquidAI/LFM2-8B-A1B

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🌟 LiquidAI/LFM2-8B-A1B 📄

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🔗 Text Generation

🔥 Transformers

🛡 Safetensors

🌐 8 languages

lfm2_moe

liquid

lfm2

edge

🧠 Mixture of Experts

conversational

custom_code

📄 License: lfm1.0

📄 Model card

📁 Files and versions 🔗 xet

👤 Community 4

📘

🔍 Train

🚀 Deploy



LFM2-8B-A1B

LFM2 is a new generation of hybrid models developed by [Liquid AI](#), specifically designed for edge AI and on-device deployment. It sets a new standard in terms of quality, speed, and memory efficiency.

We're releasing the weights of our first MoE based on LFM2, with 8.3B total parameters and 1.5B active parameters.

- LFM2-8B-A1B is the best on-device MoE in terms of both **quality** (comparable to 3-4B dense models) and **speed** (faster than Qwen3-1.7B).
- **Code and knowledge** capabilities are significantly improved compared to LFM2-2.6B.
- Quantized variants fit comfortably on high-end **phones, tablets, and laptops**.

Find more information about LFM2-8B-A1B in our [blog post](#).

Model details

Due to their small size, we recommend **fine-tuning LFM2 models on narrow use cases** to maximize performance. They are particularly suited for agentic tasks, data extraction, RAG, creative writing, and multi-turn conversations. However, we do not recommend using them for tasks that are knowledge-intensive or require programming skills.

Property	LFM2-8B-A1B
Total parameters	8.3B
Active parameters	1.5B
Layers	24 (18 conv + 6 attn)
Context length	32,768 tokens
Vocabulary size	65,536
Training precision	Mixed BF16/FP8
Training budget	12 trillion tokens
License	LFM Open License v1.0

Supported languages: English, Arabic, Chinese, French, German, Japanese, Korean, and Spanish.

Generation parameters: We recommend the following parameters:

- temperature=0.3
- min_p=0.15
- repetition_penalty=1.05

Downloads last month

🛡 Safetensors ①

Model size 8.34B params

Tensor type F32

🗨 Chat template

📄 Files info

🔗 Inference Providers NEW

🔗 Text Generation

This model isn't deployed by any Inference Provider.

🔗 2 Ask for provider support

🌳 Model tree for LiquidAI/LFM2-8B-A1B ①

🔗 Finetunes 2 models

🔗 Quantizations 15 models

🗂 Spaces using LiquidAI/LFM2-8B-A1B 2

🔗 Luigi/ZeroGPU-LLM-Inference

🔗 Mohaddz/LFM2-8B-A1B

📁 Collection including LiquidAI/LFM2-8B-A1B

🔗 LFM2 📁 Collection

LFM2 is a new generation of hybrid model... • 21 items • Updated 1 day ago • 🔗 107



LFM2 - Liquid Foundation Model: Fast On-Device AI: Install and Test Locally

Fahd Mirza • 2.4K views • 2 months ago

This video locally installs LFM2, specifically designed for edge AI and on-device deployment. 🌟 Get 50% Discount on any A6000 or A5000 GPU rental, use following link and coupon: <https://bit.ly...>



LFM2 Audio: End-to-End Audio Foundation Model: Install and Test Locally

Fahd Mirza • 1.3K views • 5 days ago

This video locally installs and tests LFM2-Audio-1.5B which is end-to-end audio foundation model. 🌟 Access ChatGPT 5, Perplexity, Claude and 500 more LLMs & tools with a single subscription...



LFM2-VL 450M: Tiny AI for Images with Variable Resolutions: Install Locally

Fahd Mirza • 1.9K views • 1 month ago

This video locally installs and tests LFM2-VL, designed to process text and images with variable resolutions. 🌟 Buy Me a Coffee to support the channel: <https://ko-fi.com/fahdmirza>

```
(ai) Ubuntu@0360-kci2-ty6k-prxmx100037:~$ pip install torch torchvision torchaudio --index-url https://download.pytorch.org/whl/cu121
pip install git+https://github.com/huggingface/transformers
pip install git+https://github.com/huggingface/accelerate
pip install huggingface_hub
Looking in indexes: https://download.pytorch.org/whl/cu121
Collecting torch
  Using cached https://download.pytorch.org/whl/cu121/torch-2.5.1%2Bcu121-cp311-cp311-linux_x86_64.whl (780.5 MB)
```

```
Using cached https://download.pytorch.org/whl/nvidia_nvjitlink_cu12-12.9.86-py3-none-manylinux2010_x86_64.manylinux2_12_x86_64.whl (39.7 MB)
Installing collected packages: mpmath, typing-extensions, sympy, pillow, nvidia-nvtx-cu12, nvidia-nvjitlink-cu12, nvidia-nccl-cu12, nvidia-curand-cu12, nvidia-cufft-cu12, nvidia-cuda-runtime-cu12, nvidia-cuda-nvrtc-cu12, nvidia-cuda-cupti-cu12, nvidia-cublas-cu12, numpy, networkx, MarkupSafe, fsspec, filelock, triton, nvidia-cusparse-cu12, nvidia-cudnn-cu12, Jinja2, nvidia-cusolver-cu12, torch, torchvision, torchaudio
13/26 [numpy]
```

```
23.0->huggingface hub) (3.10)
Requirement already satisfied: h11>=0.16 in ./miniconda3/envs/ai/lib/python3.11/site-packages (from httpcore==1.*->httpx<1,>=0.23.0->huggingface hub) (0.16.0)
Requirement already satisfied: sniffio>=1.1 in ./miniconda3/envs/ai/lib/python3.11/site-packages (from anyio->httpx<1,>=0.23.0->huggingface hub) (1.3.1)
Requirement already satisfied: click>=8.0.0 in ./miniconda3/envs/ai/lib/python3.11/site-packages (from typer-slim->huggingface hub) (8.3.0)
(ai) Ubuntu@0360-kci2-ty6k-prxmx100037:~$
conda install -c conda-forge --override-channels notebook -y
conda install -c conda-forge --override-channels ipywidgets -y
jupyter notebook
Channels:
- conda-forge
Platform: linux-64
Collecting package metadata (repodata.json): /
```

```
localhost:8888/notebooks/Untitled14.ipynb 150%
jupyter Untitled14 Last Checkpoint: 1 minute ago
File Edit View Run Kernel Settings Help Trusted
JupyterLab Python 3 (ipykernel)

[1]: from transformers import AutoModelForCausalLM, AutoTokenizer

model_id = "LiquidAI/LFM2-8B-A1B"
model = AutoModelForCausalLM.from_pretrained(model_id, device_map="auto", dtype="bfloat16",)
tokenizer = AutoTokenizer.from_pretrained(model_id)

config.json: 1.22k? [00:00<00:00, 153kB/s]
model.safetensors.index.json: 210k? [00:00<00:00, 32.3MB/s]
Fetching 7 files: 100% 7/7 [00:50<00:00, 50.30s/it]
model-00004-of-00007.safetensors: 100% 5.36G/5.36G [00:44<00:00, 100MB/s]
model-00005-of-00007.safetensors: 100% 5.36G/5.36G [00:46<00:00, 87.5MB/s]
model-00003-of-00007.safetensors: 100% 5.36G/5.36G [00:49<00:00, 320MB/s]
model-00001-of-00007.safetensors: 100% 5.35G/5.35G [00:50<00:00, 205MB/s]
model-00007-of-00007.safetensors: 100% 1.19G/1.19G [00:10<00:00, 156MB/s]
model-00006-of-00007.safetensors: 100% 5.37G/5.37G [00:48<00:00, 183MB/s]
```

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jupyter Untitled14 Last Checkpoint: 1 minute ago
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JupyterLab Python 3 (ipykernel)

[2]: prompt = """
Hey smile, you are on camera. I am recording this for a youtube video.
"""

input_ids = tokenizer.apply_chat_template([{"role": "user", "content":prompt}],add_generation_prompt=True,return_tensors="pt",tokenize=True,)

output = model.generate(input_ids, do_sample=True, temperature=0.3, min_p=0.15, repetition_penalty=1.05, max_new_tokens=512,)

print(tokenizer.decode(output[0], skip_special_tokens=False))

<|startoftext|><|im_start|>user

Hey smile, you are on camera. I am recording this for a youtube video.
<|im_end|>
<|im_start|>assistant
Hello! 😊 Smile back—I'm here and ready to chat! Great to be on camera with you. This sounds like a fun video—what's the vibe? Whether it's
casual conversation, tips, or something creative, I'm all in! Let's make it engaging! ✨<|im_end|>

[ ]:
```

Ubuntu@0360-kcf2-ty6k-prmx100037: ~

Device 0 [NVIDIA RTX 6000 Ada Generation] PCIe GEN 4@16x RX: 83.00 KiB/s TX: 15.00 KiB/s
GPU 2505MHz MEM 9500MHz TEMP 41°C FAN 30% POW 86 / 300 W
GPU[|||||||] 23% MEM[|||||||] 16.700Gi/47.988Gi



PID	USER	DEV	TYPE	GPU	GPU MEM	CPU	HOST MEM	Command
1134167	Ubuntu	0	Compute	23%	16566MiB	34%	100%	1499MiB /home/Ubuntu/miniconda3/envs/ai/bin/python -m ip
1193	root	0	Graphic	0%	10MiB	0%	0%	12MiB /usr/lib/xorg/Xorg vt1 -displayfd 3 -auth /run/u
2022	gdm	0	Graphic	0%	8MiB	0%	0%	48MiB /usr/bin/gnome-shell

F2Setup F6Sort F9Kill F10Quit F12Save Config

3 (ipykernel) Trusted

chat_template.jinja: 1.30k/? [00:00<00:00, 255kB/s]

```
[3]: prompt = """
Write a short anime story set in a futuristic Tokyo where neon dragons fly through the skyline and music controls reality itself.
The main character, Aiko, is a teenage DJ who discovers her beats can bend time. Include emotion, action, and a touching ending
that feels like the final episode of a powerful anime series.
"""

input_ids = tokenizer.apply_chat_template([{"role": "user", "content": prompt}], add_generation_prompt=True, return_tensors="pt", tokenize=True)
im_end_id = tokenizer.convert_tokens_to_ids("<|im_end|>")
eos_id = None if im_end_id == tokenizer.unk_token_id else im_end_id
output = model.generate(input_ids, do_sample=True, temperature=0.3, min_p=0.15, repetition_penalty=1.05, max_new_tokens=2512, eos_token_id=eos_id)
print(tokenizer.decode(output[0, input_ids.shape[-1]:], skip_special_tokens=True).strip())
```

Hello! 😊 Smile back—I'm here and ready to record! This is going to be a fun video! Let's make it engaging and authentic. What's the vibe yo

localhost:8888/notebooks/Untitled14.ipynb 150%

jupyter Untitled14 Last Checkpoint: 9 minutes ago

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JupyterLab Python 3 (ipykernel)

```
***Title: *Echoes of the Neon Dragon***

In the shimmering sprawl of Neo-Tokyo, where skyscrapers pulsed with neon light and the air hummed with electric energy, music wasn't just sound—it was magic. Every beat, every note, shaped reality. The city's heartbeat was its soundtrack, and those who could control it—DJs like Aiko—were legends.

Aiko stood on the rooftop of her family's old synth shop, breath misting in the cool night air. Her fingers danced over a custom-built turntable, syncing with a hidden rhythm only she could hear. The sky above roared alive—neon dragons soared, their scales glowing like liquid light, each wingbeat sending ripples through time.

Tonight, she'd push further.

She pressed play.

A low, resonant bass pulsed beneath her feet. The dragon *Shinigami*, a crimson beast with wings like storm clouds, materialized beside her, eyes glowing with ancient wisdom. "You're ready," it rumbled, voice like thunder.

Aiko smiled, heart racing. "Let's make something beautiful."

With a twist of her mix, the city shifted. Buildings shimmered, memories flickered like old film reels, and time bent—moments stretched, paused, replayed. She spun a beat so pure, so full of longing, that the world held its breath. A child laughed, a flower bloomed mid-air, and a forgotten dream rose from the past.

But the power came at a cost. Each use drained her strength, each note a thread pulled from the fabric of time. Shinigami's voice softened. "Every beat costs you something. Remember—music shapes reality, but reality shapes your soul."

Aiko nodded, tears glistening. "I won't lose myself."
```

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JupyterLab Python 3 (ipykernel)

```
eyes glowing with ancient wisdom. "You're ready," it rumbled, voice like thunder.

Aiko smiled, heart racing. "Let's make something beautiful."

With a twist of her mix, the city shifted. Buildings shimmered, memories flickered like old film reels, and time bent—moments stretched, paused, replayed. She spun a beat so pure, so full of longing, that the world held its breath. A child laughed, a flower bloomed mid-air, and a forgotten dream rose from the past.

But the power came at a cost. Each use drained her strength, each note a thread pulled from the fabric of time. Shinigami's voice softened. "Every beat costs you something. Remember—music shapes reality, but reality shapes your soul."

Aiko nodded, tears glistening. "I won't lose myself."

The dragon's wings flared, a final crescendo erupting across the sky. The neon dragons swirled in a spiral of light, and for a moment, time stood still—then, with a sigh, the world resumed.

The city breathed again.

Aiko stepped back, exhausted but radiant. Shinigami faded into stardust, leaving behind a single note—a shimmering fragment that floated upward, dissolving into the sky.

That night, as Tokyo slept beneath a blanket of stars, Aiko played one last track. The neon dragons danced above, their wings painting constellations in sound. And somewhere, in the quiet pulse of the city, a new beat began—soft, steady, eternal.

*This is not the end.*

It's the beginning.

And in the heart of Neo-Tokyo, music still heals.
```

```
special_tokens_map.json: 100% 434/434 [00:00<00:00, 103kB/s]
chat_template.jinja: 1.30k/? [00:00<00:00, 255kB/s]

•[4]: prompt = """
Translate the following English sentence into the following languages: English, Arabic, Chinese, French, German, Japanese, Korean, and Spanish.

Sentence: "The light of hope shines in every heart."

Please format your response as:
Arabic: ...
Chinese: ...
French: ...
German: ...
Japanese: ...
Korean: ...
Spanish: ...
Swedish: ...
Indonesian: ...
Hindi: ...
Urdu: ...
"""

input_ids = tokenizer.apply_chat_template([{"role": "user", "content": prompt}], add_generation_prompt=True, return_tensors="pt", tokenize=True)
im_end_id = tokenizer.convert_tokens_to_ids("<|im_end|>")
eos_id = None if im_end_id == tokenizer.unk_token_id else im_end_id
output = model.generate(input_ids, do_sample=True, temperature=0.3, min_p=0.15, repetition_penalty=1.05, max_new_tokens=2512, eos_token_id=eos_id)
print(tokenizer.decode(output[0, input_ids.shape[-1]:], skip_special_tokens=True).strip())
```

```
input_ids = tokenizer.apply_chat_template([{"role": "user", "content": prompt}], add_generation_prompt=True, return_tensors="pt", tokenize=True)
im_end_id = tokenizer.convert_tokens_to_ids("<|im_end|>")
eos_id = None if im_end_id == tokenizer.unk_token_id else im_end_id
output = model.generate(input_ids, do_sample=True, temperature=0.3, min_p=0.15, repetition_penalty=1.05, max_new_tokens=2512, eos_token_id=eos_id)
print(tokenizer.decode(output[0, input_ids.shape[-1]:], skip_special_tokens=True).strip())

English: The light of hope shines in every heart.
Arabic: نور الأمل يشرق في كل قلب.
Chinese: 希望之光照亮每一个心。
French: La lumière de l'espoir brille dans chaque cœur.
German: Das Licht der Hoffnung scheint in jedem Herzen.
Japanese: 希望の光はすべての心に輝いています。
Korean: 희망의 빛이 모든 마음속에 비춥니다.
Spanish: La luz de la esperanza brilla en cada corazón.
Swedish: Lichten förhoppnet lysar i varje hjärta.
Indonesian: Cahaya harapan terang dalam setiap hati.
Hindi: आशा की रोशनी हर हृदय में आभर्य होती है।
Urdu: آمل کی شکار دلوں میں کل جانب میں روشنی ہے۔

[ ]:
```

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JupyterLab Python 3 (ipykernel)

chat_template.jinja: 1.30k/? [00:00<00:00, 255kB/s]

```
[*]: prompt = """
Create a SINGLE, self-contained HTML file (no external assets, CDNs, or frameworks) named index.html that showcases a cool, responsive UI.

Requirements:
- Inline <style> and <script> only.
- Modern, clean layout with: a sticky header, a collapsible sidebar, a responsive card grid, a modal dialog, and a dark/light theme toggle.
- Fully responsive from 320px to desktop (use CSS Grid/Flex, no libraries).
- Accessible: semantic HTML5 tags, aria-* where helpful, focus states, Escape to close modal, keyboard navigation for the sidebar and modal,
- Smooth but subtle transitions (hover/focus/expand).
- No external fonts/images; generate simple SVG icons inline.
- Include brief comments explaining key sections.

Data:
- Generate 8-10 sample "projects" with title, tag, and short description to populate the cards.

Behavior:
- Sidebar can collapse/expand (remember state in localStorage).
- Theme toggle persists via localStorage.
- Clicking a card opens the modal with more details; modal is closable by button, overlay click, or Escape.

Output:
- Return ONLY the complete HTML file content, starting with <!doctype html> and nothing else.
"""
```

